

**Goal.** Apply calculus to parametric equations

1. Write a parametric equation for a line of slope  $\frac{4}{3}$  going through point  $(-1, 2)$ .
2. Write a parametric equation for a circle of radius 10 with center  $(4, -3)$ .

3. Consider the parametric equation  $c(t) = (x(t), y(t)) = (4t - 1, 2t^3)$ .

(a) Find a formula for the slope of a tangent line to the graph (as a function of  $t$ ).  
What is the slope when  $t = 5$ ?

(b) Find the area between  $c(t)$  and the  $x$ -axis from  $t = 0$  to  $t = 3$ .

(c) What is the speed of a particle traveling along  $c(t)$  for any value of  $t$ ? What is the speed at  $t = 0$ ?

(d) Set up but do not evaluate an integral to find the arc length of  $c(t)$  for  $-5 \leq t \leq 5$ .

(e) Find the surface area of the surface of revolution created by revolving  $c(t)$  for  $0 \leq t \leq 2$  about the  $x$ -axis.